Appl Phys A (2008) 93: 955–959
DOI 10.1007/s00339-008-4773-y

Applied Physics A
Materials Science & Processing

Ultrafast laser based "green" synthesis of non-toxic nanoparticles in aqueous solutions

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Received: 12 October 2007 / Accepted: 9 April 2008 / Published online: 4 July 2008
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Abstract Inorganic nanoparticles offer novel promising properties for biological sensing and imaging, as well as in therapeutics. However, these applications are often complicated by the possible toxicity of conventional nanomaterials, arising as a result of inadequate purification procedures of nanoparticles obtained via synthetic pathways using toxic or non-biocompatible substances. We review novel femtosecond laser-assisted methods, which enable the preparation of metal nanomaterials in clean, biologically friendly aqueous environment ("green" synthesis) and thus completely solve the toxicity problem. The proposed methods, including laser ablation and fragmentation, make possible the production of stable metal colloids of extremely small size (~2 nm) with a low coefficient of variation (15–25%). Those nanoparticles exhibit unique surface chemistry and can be used for bio-imaging, cancer treatment and nanoparticle-enhanced Raman spectroscopy.

PACS 81.16.Mk · 81.05.-t

1 Introduction

With the synthesis of new inorganic nanomaterials, nanobiophotonics is the focus of interest of numerous research groups, promising many important applications in medicine, food and water quality, defense and security, etc. [1]. In all cases, newly synthesized nanomaterials are expected not only to exhibit superior properties compared to traditional organic materials, but also to bring novel and unique characteristics for biosensing and imaging. In particular, semiconductor-based quantum dots (QDs) can provide much better photoluminescence properties compared to organics fluorophores, which suffer from optical bleaching, broad and asymmetric emission spectra, single wavelength monomodal imaging, etc. [2]. Furthermore, noble metal nanostructures can support localized plasmons, coupled to collective excitations of electrons [3, 4], giving rise to a strong resonant absorption and scattering at a particular wavelength (the scattering cross section of gold exceeds that of dielectric nanoparticles by several orders of magnitude [5]) and an extremely strong amplification of the electric field near the nanoparticle [6]. All these effects promise a successful use of plasmonic structures as contrast agents in imaging [7], as photosensitizers in therapeutics [5, 8], as well as agents for Raman scattering amplification [9].

Despite the above-stated properties of inorganic nanomaterials, their use in many applications (especially in vivo) remains under scrutiny. The main problem is related to their possible toxicity and/or incompatibility with biological objects or tissues [10]. In fact, one can distinguish two potential toxicity problems. The first one is related to the inherent toxicity of some chemical compounds forming the nanoparticle (e.g., surfactants) or released from nanoparticles interacting with the biological environment. For example, certain Cd-based quantum dots have been shown to release Cd^{2+} ions in the cellular compartment during in vitro imaging by confocal laser fluorescence microscopy [2, 11–13]. Unfortunately, this kind of toxicity is difficult to overcome. Although it can somehow be reduced by an encapsu-

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lation of the potentially toxic compounds, the long-term stability of such complexes remains of great concern and uncertainty, which almost prevents their approval by the Food and Drug Administration (FDA). The second toxicity is related to the use of toxic or non-biocompatible compounds during the chemical synthesis procedure. In particular, conventional methods for the fabrication of colloidal metal nanostructures (nanoparticles, nanorods, core-shells) are based on a chemical reduction process [14] which generally gives rise to the surface contamination by residual anions and reducing agents [14–18]. The resulting toxicity of such nanostructures complicates their applications in in vivo biosensing and imaging [10].

To solve the toxicity problem, we recently introduced ultrashort laser ablation in aqueous biocompatible solutions as a novel method to synthesize non-toxic nanomaterials [19–27]. Such ablation of materials leads to the formation of stable colloids, whereby parameters such as size and size distribution can be controlled by adjusting the laser parameters [19] and by adding capping ligands in the solution [20–24]. Furthermore, the fabrication of nanoparticles of narrow size distribution can be achieved even in deionized water when no reactive species are present in the solution [19, 25, 26]. Such "green" synthesis in biologically friendly environment enables the production of uncontaminated and non-toxic particles due to the absence of toxic chemical reducing agents and capping ligands.

In this paper, we review our recent progress in the development of "green" femtosecond laser-based nanofabrication in liquid, which includes both the laser ablation from a solid target and the fragmentation of already formed colloids.

2 Femtosecond laser ablation from a solid target in aqueous solutions

In this approach, femtosecond laser radiation is used to ablate a solid target (Au, Ag, etc.) placed on the bottom of a glass vial filled with deionized water or various biocompatible aqueous solutions. In particular, in our experiments we used a Hurricane Ti:Sapphire femtosecond laser (Spectra Physics) emitting 120 fs pulses centered at 800 nm with a maximum energy of 1 mJ/pulse at a repetition rate of 1 kHz [19]. The laser energy was controlled by a set of fixed and variable attenuators and the spot size was adjusted to be 250 μm by a motorized microscope objective. The target is normally moved in the focusing plane using a motorized stage to obtain identical surface condition during the ablation process. In the case of Au or Ag target, such ablation process normally leads to a coloration of the solution in the vial within a few seconds after the beginning of the ablation process (the solution becomes red and yellow in the case of Au and Ag, respectively). Such coloration is related to the excitation of surface plasmons of newly formed colloidal metal nanoparticles.

Using femtosecond laser ablation, it is possible to tune the size distribution of the nanoparticles by changing radiation parameters [19]. In particular, fine control of the mean size of nanoparticles from 100 to 4 nm can be achieved by varying the laser energy from 500 to 30 μJ (Fig. 1(a)). However, the small size and size dispersion obtained at ultra-low laser energies were compromized by an extremely low production rate. As shown in Fig. 1(b), the decrease of the energy from 500 to 90 μJ led to the decrease of the production rate from 0.14 to 0.01 mg/min. Below 90 μJ the production rate was extremely low, requiring 6 to 8 hours to produce a fraction of a milligram.

The addition of capping ligands in the aqueous medium during the laser ablation process enabled the production of small and low size-dispersed nanoparticles at a high production rate. We were the first to introduce biocompatible ligands, including oligosaccharides (Cyclodextrins) and biopolymers (dextran, poly(ethylene) glycol (PEG), chitosan, etc.), which enable an extremely effective nanoparticle size control. As shown in Fig. 2, for a high laser energy of 100 μJ ensuring a relatively high production rate, the mean sizes of the resulting nanoparticles were 2.5, 3.2, and 7.7 nm, for dextran (MW = 6000 Daltons, 2g/L), β-cyclodextrin (15 mM), and PEG (MW = 2000 Daltons, 2g/L), respectively. Such a strong reduction of the nanoparticle size in

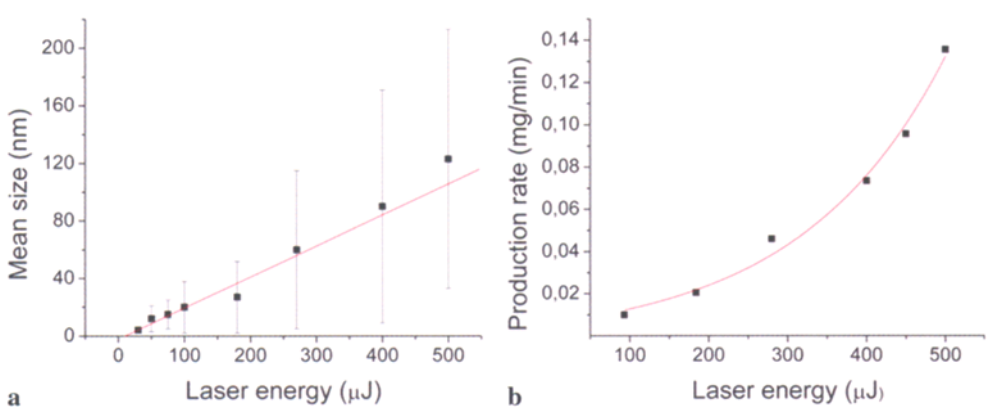

Fig. 1 (**a**) Size characteristics of nanoparticles produced in deionized water and (**b**) the production rate as a function of the laser energy used during the laser ablation. The *error bars* in (**a**) correspond to the measured dispersions

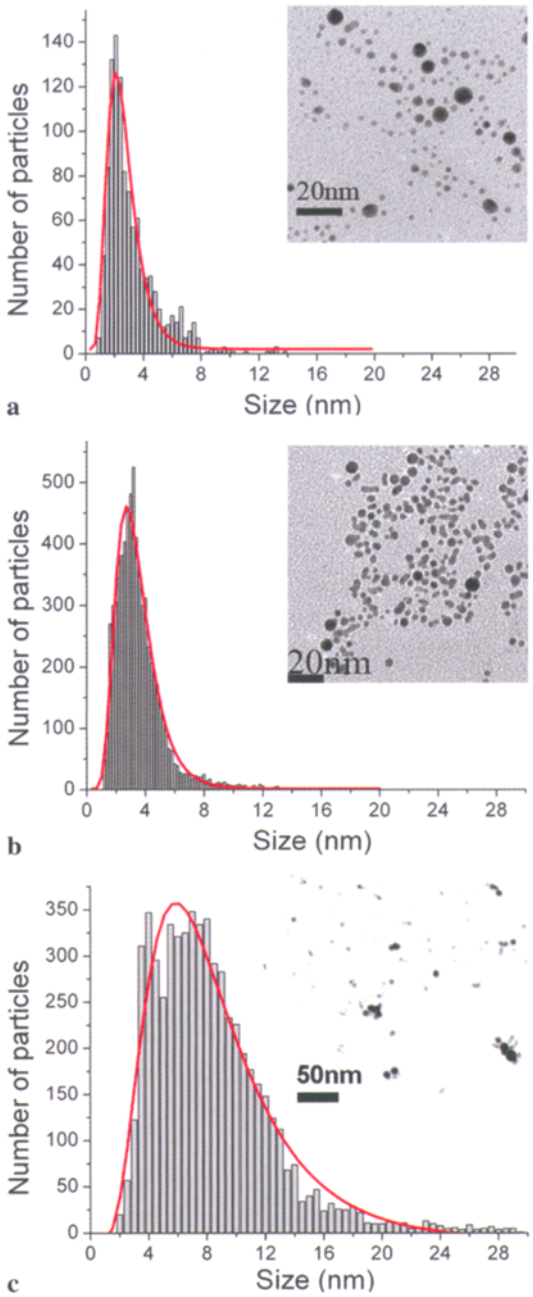

Fig. 2 Gold nanoparticles produced by laser ablation in aqueous solution of (**a**) dextran (MW = 6000 Daltons, 2g/L), (**b**) β-cyclodextrin (15 mM), and (**c**) PEG (MW = 2000 Daltons, 2g/L)

the presence of oligosaccharides or biopolymers was explained by the unique surface chemistry of laser-synthesized nanoparticles [21, 22]. Indeed, due to a partial oxidation and to the formation of O^- groups on their surface, gold nanoparticles are able to react with OH groups of different biocompatible substances [21], which usually do not take place in the absence of suitable reagents. Another advantage consists in the possibility of further nanoparticle functionalization by a desirable chemical group (e.g., thiol, amine, or hydro carboxyl) during the ablation process. The latter property is very important for projected biosensing and imaging applications.

3 Femtosecond laser fragmentation of nanoparticles

This newly introduced approach enables one to efficiently control the size distribution and production rate of nanoparticles in neutral (deionized water) or in a reactive environment [25, 26]. In particular, for the neutral environment the proposed methodology consists of two steps. The first step uses a rapid production, at high laser energies, of colloids of wide size distribution. Then, during a second step the solution is transferred in another glass vessel and illuminated by a smoothly focused (down to a spot size of 250 to 500 µm) femtosecond laser beam. The strong non linear interaction between the femtosecond laser pulse and the water medium dramatically broadens the spectral distribution of the initial pulse, resulting in a white light supercontinuum. As we showed in [25], the generation of supercontinuum strongly enhances the energy transfer to the nanoparticle and is mainly responsible for the fragmentation of the nanoparticles in the solution.

The final size of nanoparticles is independent of the initial characteristics of the colloids and is mainly determined by the laser energy and the presence of capping ligands during the fragmentation step [25, 26]. In the absence of capping ligands, the obtained nanoparticles are rather large, with a mean size ranging from 20 to 40 nm. A typical size distribution for nanoparticles produced by the laser fragmentation in pure water is shown in Fig. 3(a). Adding capping ligands to the solution drastically reduced the final nanoparticle size. As shown in Fig. 3(b), the nanoparticles mean size dropped below 10 nm when dextran (MW = 40000 Daltons, 1g/L) was added to the solution. In both cases, the size distributions of the resulting colloids were remarkably narrow, promising a successful use of nanoparticles in biosensing and imaging tasks.

4 Comparison of ultrafast laser based methods with chemistry routes

A comparison of size characteristics of nanoparticles produced by wet chemistry and by the ultrafast laser ablation-based technique is schematically given in Fig. 4. The best results in term of size selectivity and monodispersity are obtained through the Brust–Shiffrin [28–30] (2-phases) and the Turkevitch [31, 32] (water) synthesis. Using these techniques, nanoparticles with a coefficient of variation (COV) smaller than 10% can be achieved and the mean size can be easily tuned between 1 and 150 nm simply by varying

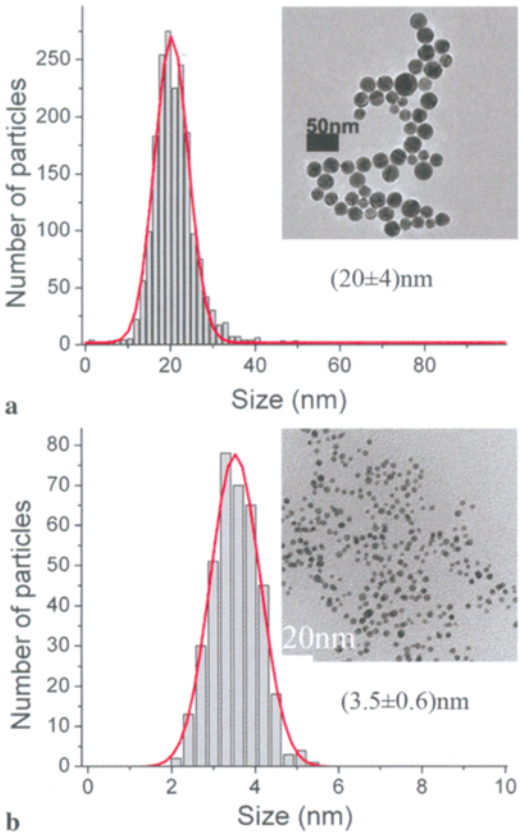

Fig. 3 Gold nanoparticles produced by the two-step (ablation and fragmentation) laser synthesis in (**a**) water and (**b**) aqueous solution of dextran (MW = 40000 Daltons, 1g/L)

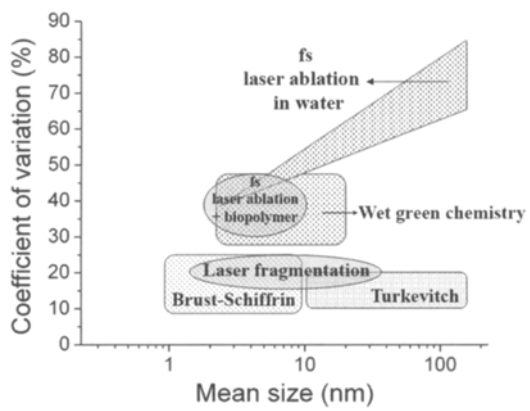

Fig. 4 Comparison of the size characteristics of nanoparticles produced by femtosecond laser synthesis, water-based wet green chemistry and Brust–Schiffrin/Turkevitch "non-green" chemical synthesis. The size and the coefficient of variation (COV) of the laser synthesized nanoparticles significantly decrease upon the addition of capping ligands and after the laser fragmentation procedure

the relative concentration of metal salts and capping ligands (Fig. 4). However, those synthesis routes are seen as "non-green" chemistry (NGC) as they rely on toxic and highly reactive reducing agents and/or the use of organic solvent. The green chemistry is underlined by 12 principles [33] and aims to the design of safer nanomaterials, reduced environmental impact, maximisation of materials and energy efficiencies, and to the minimisation of waste [34]. Numerous groups have proposed "greener" water-based wet chemistry approaches for the synthesis of nanomaterials in which biocompatible reducing agents and capping ligands are used but size and other characteristics of these nanomaterials are still far from what one can obtain by the conventional wet synthesis [35–38]. The mean size of the nanoparticles can be controlled between few nanometers and few tens of nanometers again by varying the relative concentration of metal salts and capping agents but due to the lower binding efficiency of the capping ligands the COV usually varies between 30 and 45% (Fig. 4). Furthermore, the green chemistry is usually restricted by the small reduction potential of the reducing agent [37]. This prevents the synthesis of nanoparticles such as Fe, Co, and Ni, for which ions possess a negative reduction potential [38]. In contrast, the two-step femtosecond laser synthesis (ablation and fragmentation) is able to produce nanoparticles with much better size characteristics (2–40 nm, COV ~15–25%) in comparison to the direct laser ablation-based synthesis and competes with the traditional wet chemistry (Fig. 4). Furthermore, the laser-assisted methods minimize the risk of toxicity as they directly produce neutral atoms/nanoparticles in an essentially clean and biocompatible environment. Such unique environment is impossible to reproduce even by green wet chemistry, since some unreduced metal salts are always present in the solution. This means that laser-based synthesis can provide clear advantages over standard chemical reduction procedures.

5 Conclusion

Various femtosecond laser ablation-based methods have been introduced to synthesize nanoparticles in biocompatible solutions. The produced particles are stable, free of contamination and have unique surface chemistry. Furthermore, the proposed approaches are unique in avoiding the secondary toxicity related to the traditional chemical synthesis procedure. Therefore, it is especially effective for producing nanomaterials, which are biocompatible and non-toxic by their nature. This includes the fabrication of metal-based plasmonic nanostructures (Au, Ag, Pt), some semiconductor-based nanoparticles (Si, Ge-based quantum dots, ZnO, etc.) and magnetic nanoparticles (Ni, Co, Fe_2O_3, etc.). Unconjugated or functionalized nanoparticles can be used in biological applications such as surface-enhanced Raman scattering or in vivo imaging.

Acknowledgements The authors acknowledge the financial contribution from the Natural Science and Engineering Research Council of Canada.

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